

ROBOTICS AND
AUTOMATION

**Final Project
Demonstration**

Detachable Tool-Changing Mechanism for Unitree G1

GROUP 1

PRESENTED BY

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Outline

01

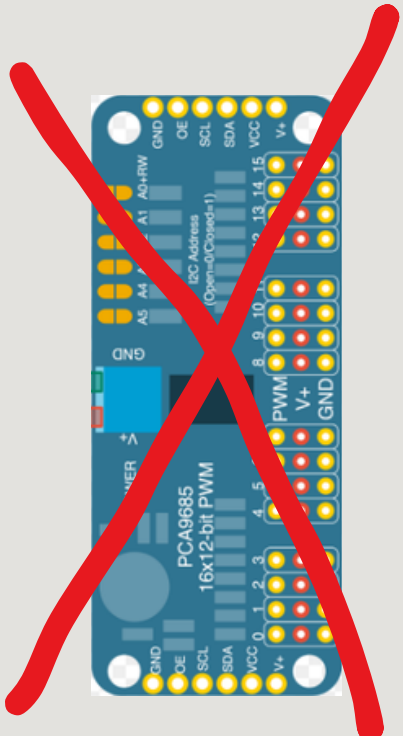
- 1. System Architecture**
- 2. Interchangable End Factor Robotic Arm**
- 3. Rotating Tool Station Modular**
- 4. Control Web Interface**
- 5. Demo**

System Architecture

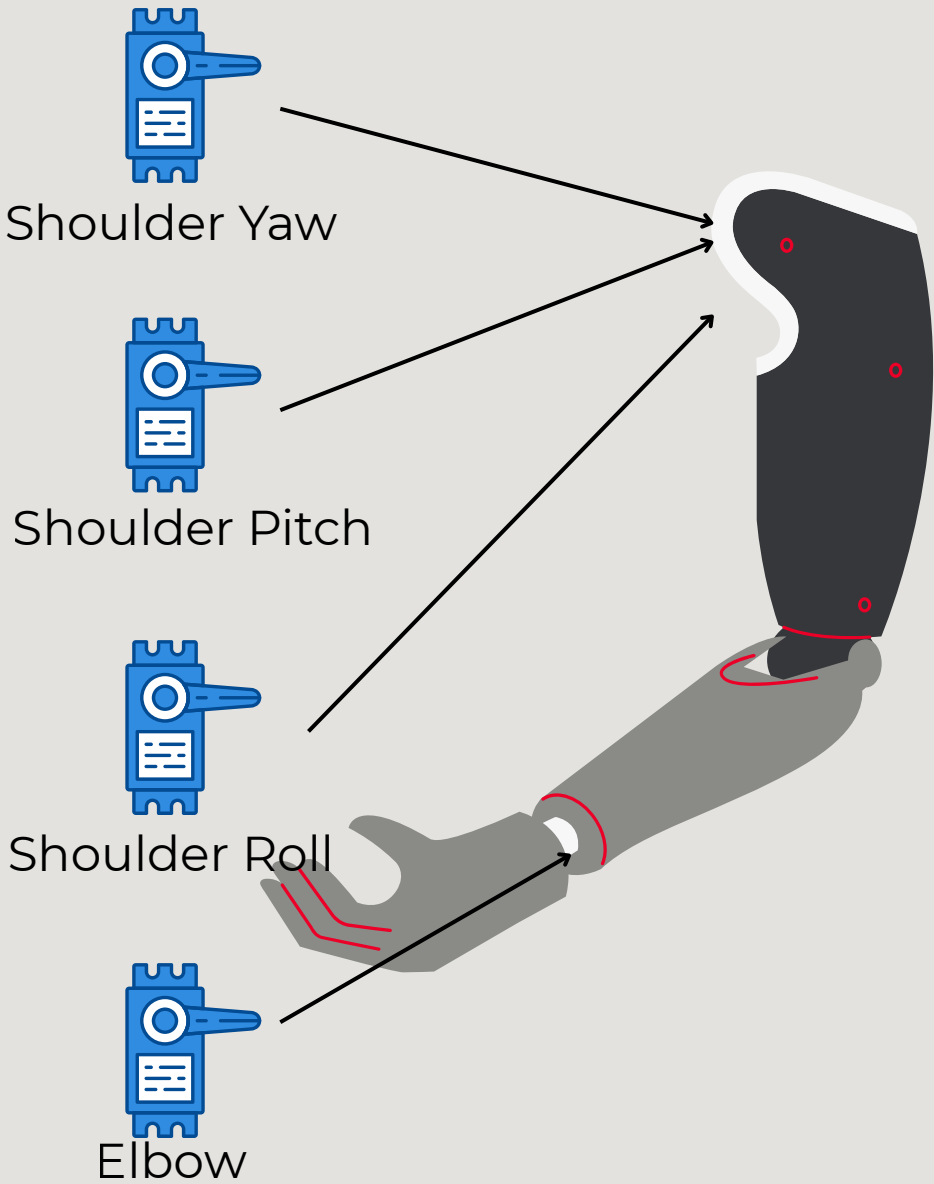
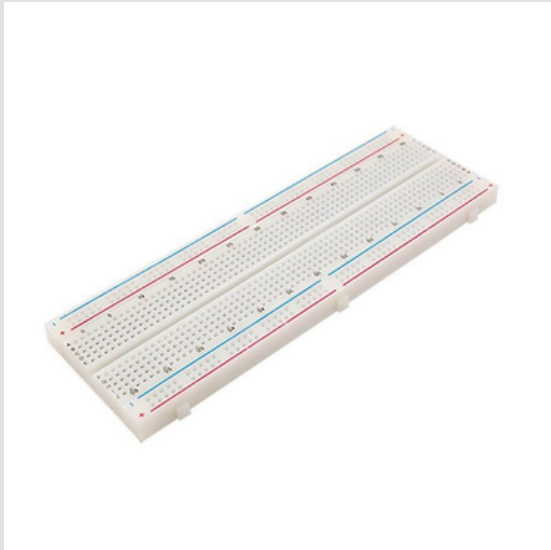
Electronical Components



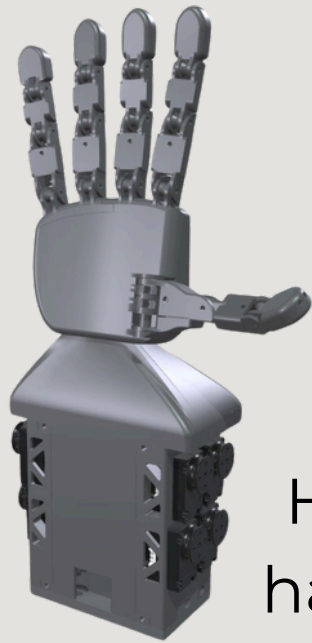
MCU



PWM Servo Driver

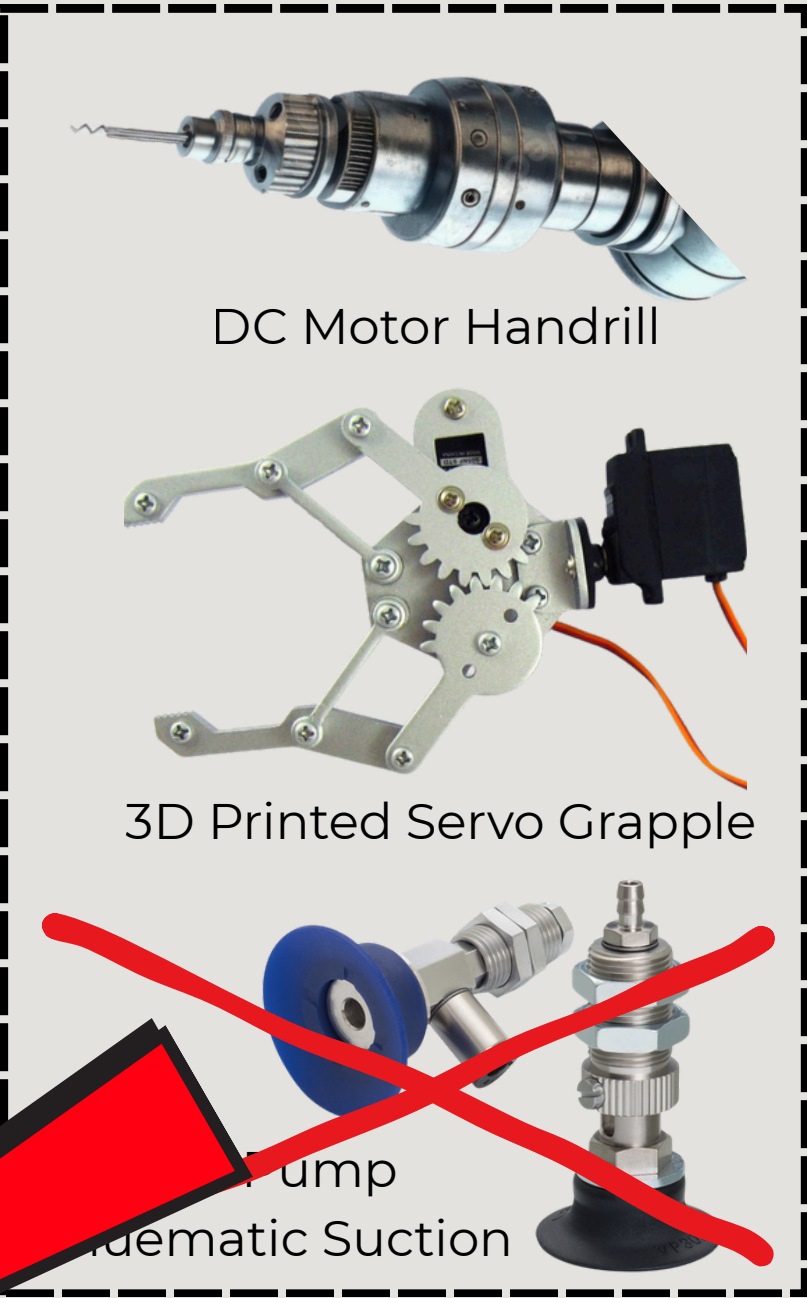


Special Coupling Mechanism



Humanoid hand model

Wrist Tools Exchanger

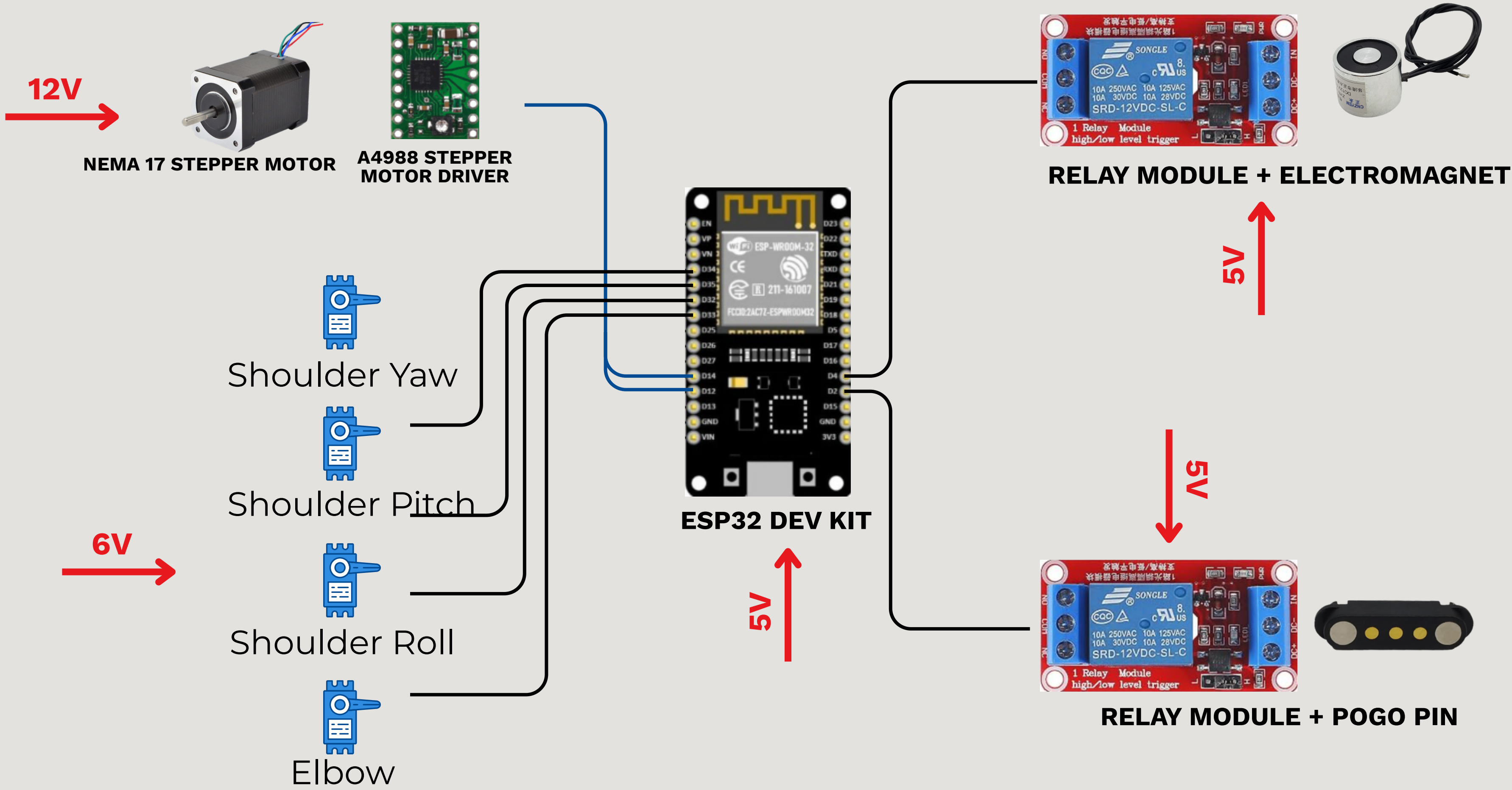


DC Motor Handrill

3D Printed Servo Grapple

Pump Automatic Suction

Power & Hardware Wiring



Interchangable End Factor Robotic Arm

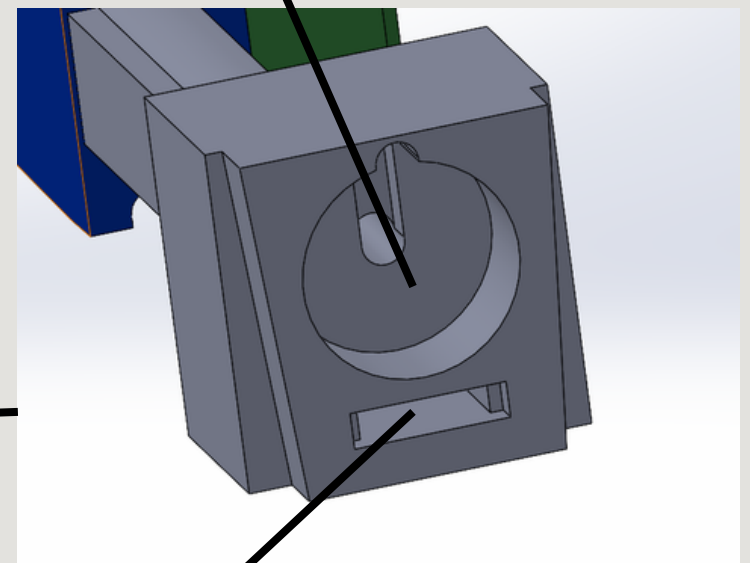
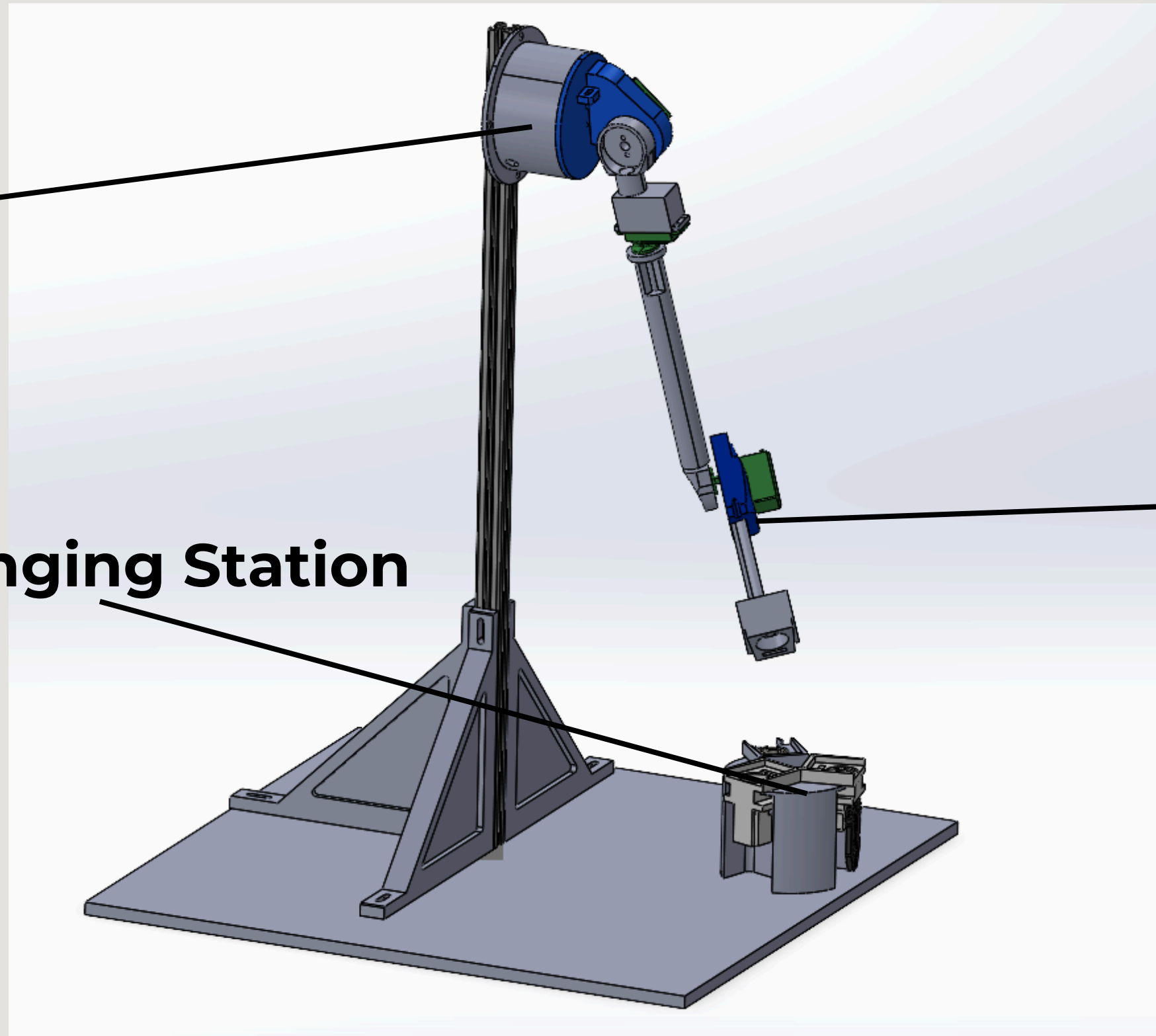
Robotic Arm

Exchanging Station

Electromagnet

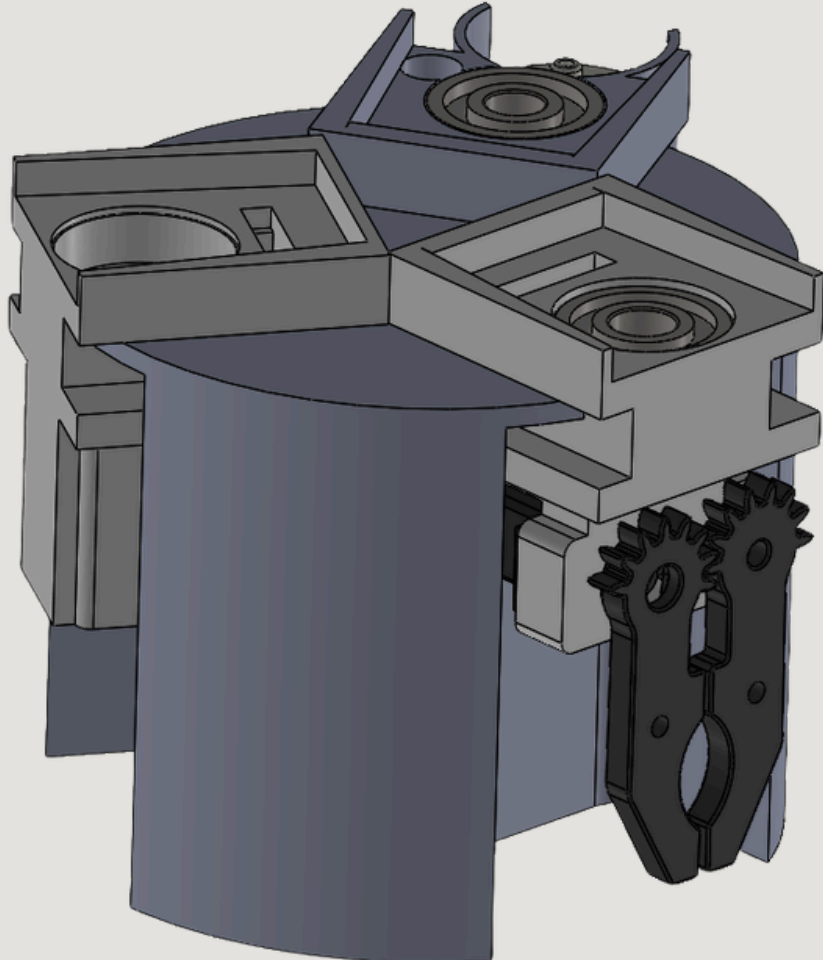
End Factor

Tool power source



Rotating Tool Station Modular

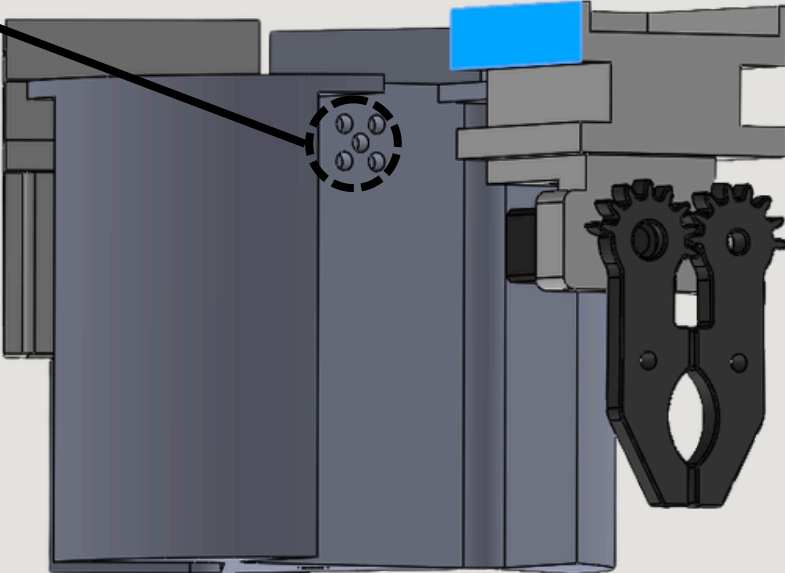
DRILL/SAW
HEAD



HAND MODLE

GRAPPLE

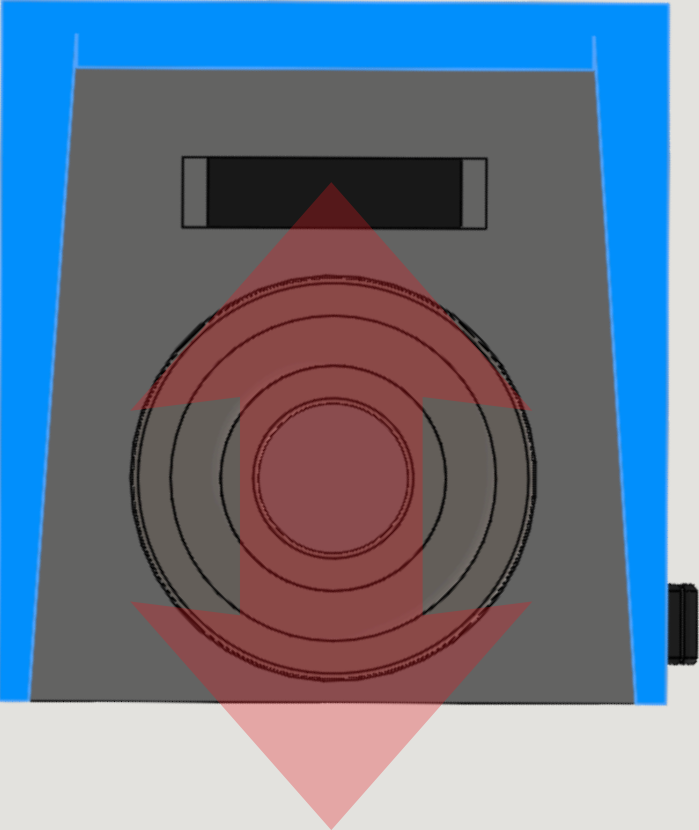
MAGNET FOR
TOOL FIXATION



NEMA 17
STEPPER
MOTOR



MECHANICAL
GUIDANCE



Control Web Interface

Connection Settings

Select COM Port

No options to select

Baud Rate

115200

Connect

Disconnect

Status: Disconnected

☒ Auto-sync status

Arm Settings

Motion Speed (ms delay)

20

Update Speed

Tool Lock / Power

Relay 1 = magnet lock. Relay 2 = pogo VCC for the current tool.

4-DOF Humanoid Arm Controller

Operator panel for arm motion, tool power, and rotary tool exchange.

System Status

HELD TOOL

None

Sync Status

STATION SLOT

2 - Gripper

Status is parsed from the ESP32 status response when synced or auto-refreshed.

MAGNET RELAY 1

OFF

TOOL POWER RELAY 2

OFF

Firmware Output

Robot Geometry Config

Tool Exchange Position

Preset Motions

Cartesian Control (IK)

Joint Control (FK)

Live Demo

Tool Station

Stepper Motor

Tool Exchange Station

The station has 3 tools on a 360 degree plate. Each slot is 120 degrees apart.

Automatic Tool Change

The robot remembers the held tool. At startup it assumes no tool is held.

Change/Pick Gripper

Change/Pick Hand

Change/Pick Drill

Initial Pickup Gripper

Initial Pickup Hand

Initial Pickup Drill

Remove Held Tool

Preset Motions

Cartesian Control (IK)

Joint Control (FK)

Live Demo

Tool Station

Stepper Motor

Cartesian Coordinate Control (IK)

Send XYZ coordinates (mm) for the end effector to reach.

X (Forward/Back)

120,00

Y (Left/Right)

160,00

Z (Up/Down)

250,00

Stepper Motor Control

Control the separated stepper motor (A4988 driver).

Degrees to move

1,00

Move Stepper

Preset Poses & Motions

Basic Poses

Rest Pose

Raise

Side

Bend

Complex Motions

Wave

Hello

Handshake

Stroke

Pick & Place

Front Pick

Run Active Tool

Tool Status

Tool Showcase Sequence

Gripper

Gripper Open

Gripper Close

Joint Offset Control

Control joint angles relative to rest pose. Rest is 0 degrees.

☐ Use number input instead of sliders

Shoulder Pitch (S0)

-3.00

Shoulder Roll (S1)

0.00

Shoulder Yaw (S2)

0.00

Elbow Pitch (S3)

Rest

Rest

Rest

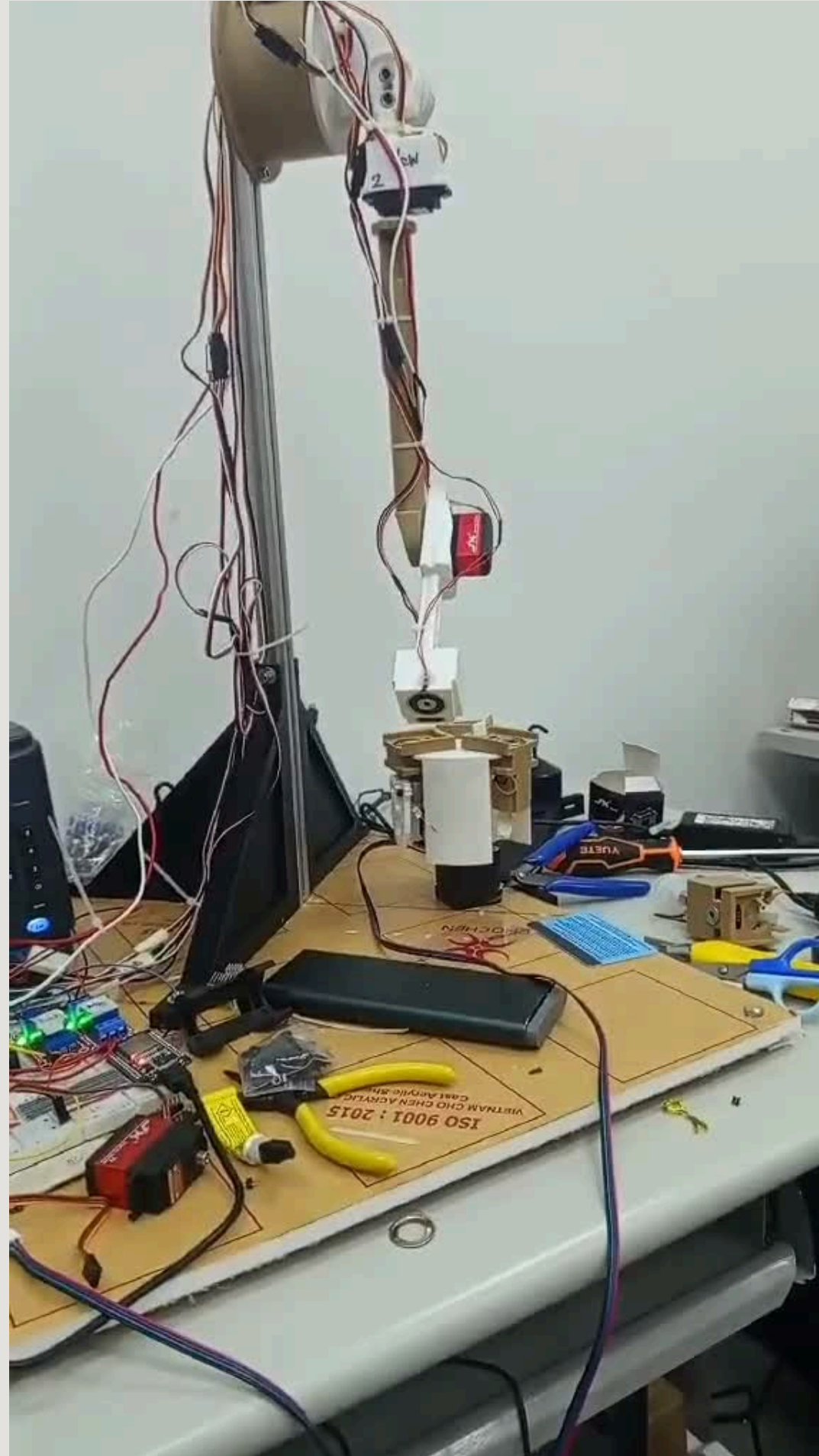
Rest

Set Joint Offsets

Rest All Joints

Run Arm Servo Diagnostic

Demo



Q&A

